

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY - LONERE.

ELECTRICAL ENGINEERING DEPARTMENT.

SUMMER EXAMINATION- APRIL - MAY, 2017.

Subject: High Voltage Engineering

Semester: VIII.

Date: 12/05/2017.

Time: 3Hr.

Marks: 70

Instruction:

- 1| Question No. 1 is compulsory.**
- 2| Solve the question in sequential manner.**
- 3| Necessary data is given wherever is required, if it is not given then it means it is part of examination.**
- 4| If data is assumed, then mention it clearly with reasons.**

Q.No.1 **Fill in the blanks with the correct answer.** **[10]**

- (i) Paper insulation is mainly used in-----application.
- (ii) The breakdown voltage of gas or air with increase in pressure under uniform fields has -----relation with pressure.
- (iii) A 16 stage impulse voltage generator has stage capacitance of $0.123 \mu\text{F}$ and charging voltage of 200 kV. The energy rating in kJ is-----.
- (iv) Typical capacitive loading on a testing transformers rated 100 kVA, 250 kV will be ---.
- (v) The best insulation among all insulating material available is-----.

Q.No.2 **Solve Any Two of the Following.** **[12]**

- (a) What is "stressed oil volume theory", how does it explain breakdown in large volume of commercial liquid dielectrics?
- (b) What will be the breakdown strength of air be for small gaps (1 mm) and large gaps (20 cm) under uniform field conditions and standard atmospheric conditions?
- (c) Discuss the various mechanisms of vacuum breakdown.

Q.No.3 **Answer Any Two of the Following.** **[12]**

- (a) What are the different methods and means for purifications of the liquid dielectrics?
- (b) What is 'thermal breakdown' in solid dielectrics, and how is it practically more significant than other mechanisms?
- (c) What are the different materials used in capacitor insulation? Discuss the insulation requirements for (i) HVDC capacitors, (ii) High rating power capacitors.

Q.No.4 Attempt Any Two of the Following.

[12]

- (a) A Cockcroft-Walton type voltage multiplier has 8-stages with capacitances, all equal to $0.05 \mu\text{F}$. The supply transformer secondary voltage is 125kV at a frequency of 150 Hz . If the load current to be supplied is 5mA , find (a) the % ripple, (b) the regulation, and (c) the optimum number of stages for minimum regulation or voltage drop.
- (b) A 12-stage impulse generator has $0.126\mu\text{F}$ capacitors. The wave front and the wave tail resistance connected are 800Ω and 5000Ω respectively. If the load capacitor is 1000pF , find the front and tail times of the impulse wave produced.
- (c) Why is a Cockcroft-Walton circuit preferred for voltage multiplier circuit? Explain its working with a schematic diagram

Q.No.5 Solve Any Two of the Following.

[12]

- (a) The HV arm of an R-C divider has 15 numbers of 120Ω resistors with a 20 pF capacitor to ground from each of the junction points. The LV arm resistance is 5Ω . Determine the capacitance needed in the LV arm for correct compensation.
- (b) Discuss the different methods of measuring high dc voltages. What are the limitations in each method?
- (c) Explain the principle and construction of an electrostatic voltmeter for very high voltages. What are its merits and demerits for high voltage ac measurements?

Q.No.6 Answer Any Two of the Following.

[12]

- (a) A 3-phase single circuit transmission line is 400 km long. If the line is rated for 220 kV and has the parameters, $R = 0.1\Omega/\text{km}$, $L = 1.26\text{ mH/km}$, $C = 0.009 \mu\text{F/km}$, and $G = 0$. Find (i) surge impedance, (ii) the velocity of propagation neglecting the resistance of the line. If a surge of 150 kV and infinitely long tail strikes at one end of the line, what is the time taken for the surge to travel to the other end of the line?
- (b) Write short notes on (i) Rod gaps used as protective devices, (ii) Ground wires for protection of overhead lines.
- (c) What is the significance of impulse test? Briefly explain the impulse testing of insulators.

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